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SrO/BaO effect on interface formation, growth and interfacial reactions between glass sealants and electrolyte for solid oxide fuel cell application

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Abstract

An appropriate glass sealant is still a herculean task to commercialize and make competitive the solid oxide fuel cell (SOFC) technology than peers. In present study, new glass sealants SrO-BaO-SiO₂-B₂O₃-ZrO₂ with variable ratio of SrO/BaO, are synthesized. The interface formation, growth and interfacial study with yttria stabilized zirconia (YSZ) is reported in the present study. The presences of single alkaline earth metal oxide like SrO or BaO in glass sealant has increased the adhesion and prevent the delamination at the interface even after exposed up to 1000 h at 900 °C in air. The mixed modifier effect also plays a very crucial role in phase formation, growth and interfacial reactions with YSZ. However, the presence of both the modifiers is increased the crystallization tendency with formation of detrimental crystalline phases. Only SrO or BaO contained glass

Outline

Abstract

Graphical abstract

Keywords

1. Introduction

2. Experimental details

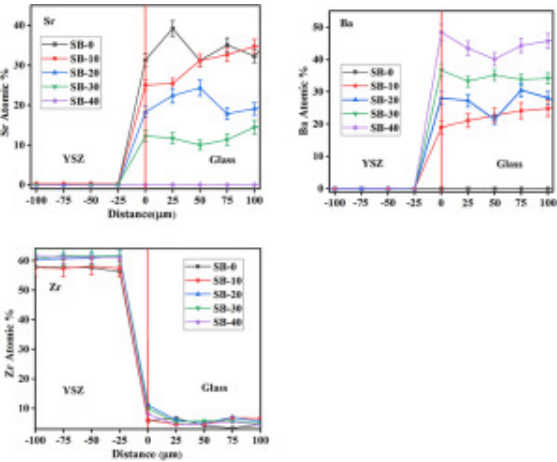
3. Results and discussion

4. Conclusion

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sealants with YSZ is formed smooth, pore free, and coherent interface with good hardness among all the developed glass sealants. These glass sealants can be tested and used as a sealant in real SOFC conditions.

Graphical abstract



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Keywords

Glass sealants; Interface; Growth; Electrolyte; SOFC

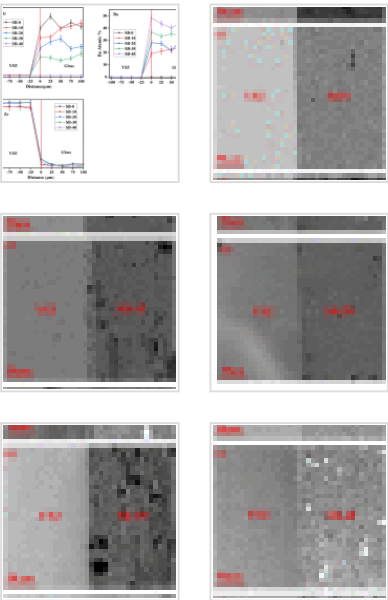
[Previous article in this issue](#)

[Next article in this issue](#)

1. Introduction

- Author statement
- CRedit authorship contribution statement
- Declaration of competing interest
- Acknowledgements
- Data availability
- References
- [Show full outline](#)

Figures (10)



[Show 4 more figures](#)

Tables (3)

[Table 1. EDS analysis of the SB-0, SB-10, ...](#)

The extensive attention has been focused to the commercialization of solid oxide fuel cell (SOFC) in recent years. It has good internal reforming ability, high-energy conversion efficiency without requiring any external catalyst like platinum etc. Moreover, hydrocarbons can also be used as fuel make them sustainable and energy sources [1]. Planar-type SOFCs, with the advantages of an easy processing and a high-power density, are considered to be superior to contemporary designs like tubular design [2]. However, commercialization of planar-type SOFCs has been hindered by insufficient long-term cell durability, caused primarily by the cracking of the glass seals among SOFC components during thermal cycling that creates passage for the fuel and oxidizing gases. Planar design of SOFC require a robust seal that can work at high working temperature of SOFC (> 800 °C) [3,4]. An ideal sealing material needs to meet several stringent requirements, notably leakage proof of gases and liquid, maintenance of mechanical integrity, and chemical, electrical, and thermal compatibility with the fuel and the SOFC components [[5], [6], [7], [8], [9], [10], [11], [12]]. A sealing glass or glass ceramic has to be thermally stable, electrically insulating capable of sustaining a coefficient of thermal expansion (CTE) between $9 - 13 \times 10^{-6} / ^\circ\text{C}$ and a viscosity is 10^5 Pas at the joining temperature and $> 10^9$ Pas at the working temperature of SOFC, mechanically strong in oxidizing, wet, and reducing atmospheres of oxygen, with a good adherence to the other components of SOFC [[13], [14], [15]].

Several glass and glass-ceramic systems, including (BaO, SrO, CaO)-Y₂O₃-B₂O₃-SiO₂ [16], (BaO, CaO)-Al₂O₃-SiO₂ [17,18] and (BaO, SrO)-Al₂O₃-La₂O₃-SiO₂-B₂O₃ [[18], [19], [20], [21], [22], [23]] have been investigated in recently for their use as sealing materials in SOFC. Extensive studies have been carried out on various glass compositions. Mostly, silicate-based glasses or borosilicate- based glasses are most common reported glass network having alkaline earth metal oxides as modifiers as sealants in SOFC. Few studies have also reported alkali metals oxides as modifiers along with some intermediate oxides like Al₂O₃, La₂O₃ and Na₂O₃ to achieve the desired properties as sealants. Further, to optimized the CTE, stability and minimize the interfacial reactions with other components, some intermediate oxides are also used in sealant [11]. Each and all glass

Table 2. Different phases with their volu...

Table 3. Diffusion constant for yttrium, s...

sealants meet some of the required properties. So, new glass sealants are being synthesized and investigated to be qualified as sealants. Singh et al. [24] has reviewed and reported new directions to develop compatible glass sealants for SOFC applications. The composition of glass sealants is selected on the basis of the common components are being used in SOFC technology such as YSZ and SrZrO_3 and BaZrO_3 , as electrolytes and cathodes, respectively. Thus, the glass compositions exhibit SrO, BaO and ZrO_2 to reduce the fast diffusion in different components of SOFC. Before using any developed sealing glasses and glass ceramics in real SOFC operating conditions. It is worthwhile to investigate their compatibility, interfacial reaction with other components of SOFC to save resources, time and avoid the failure of the device in real operating conditions. Therefore, the motivation of the present work is to study the interfacial study with well-known electrolyte materials like yttria stabilized zirconia (YSZ) for high temperature SOFC application.

Therefore, the chemical compatibility of $\text{SrO-BaO-SiO}_2\text{-B}_2\text{O}_3\text{-ZrO}_2$ glass seals with YSZ is investigated and reported in the present research work. Most importantly, the idea behind the selection of present composition is to use mixed modifiers effect to prevent excessive reactions between electrolyte (YSZ) and glass sealants [25,26]. Secondly, some quantity of ZrO_2 also include in glass composition to slow down the Zr^{4+} diffusion towards glass sealants. The mixed modifier effect on various properties is also reported in the present study. The diffusion couples of glass sealants with YSZ were heat treated at 900°C for different durations in order to understand the interface formation, growth and reactions among YSZ and developed glass sealants. These diffusion couples then characterized using X-ray diffraction (XRD), field emission scanning electron microscope (FE-SEM)-EDS and exposing Vickers Hardness and reported in the present study.

2. Experimental details

The stoichiometric amounts of SiO_2 (99 %, Loba Chemie), SrO (99 %, Loba Chemie), BaO (99 %, Loba Chemie), B_2O_3 (99 %, Loba Chemie) and ZrO_2 (97 %, Sigma Aldrich) chemicals were used for preparing the glass samples. The glass compositions $40\text{SrO-}45\text{SiO}_2\text{-}$

10B₂O₃–5ZrO₂, 30SrO–10BaO–45SiO₂–10B₂O₃–ZrO₂, 20SrO–20BaO–45SiO₂–10B₂O₃–5ZrO₂, 10SrO–30BaO–45SiO₂–10B₂O₃–5ZrO₂ and 40BaO–45SiO₂–10B₂O₃–5ZrO₂ are taken in mol % and hereafter called SB-0, SB-10, SB-20, SB-30 and SB-40, respectively. The preparation of these glasses and their thermal and structural properties were reported in our earlier published work [27,28]. The analytical composition of as prepared glasses is given in [Table 1](#). The as prepared glass sealants were ground and used for further investigation. High purity YSZ (TZ-8YS, Tosoh) is used as electrolyte to carry out the interfacial study.

Table 1. EDS analysis of the SB-0, SB-10, SB-20, SB-30 and SB-40 glasses in atomic%.

Label	Sr	Ba	Al	Si	B	Zr	O
SB-0	33.51 ± 1.15	–	8.49 ± 0.51	29.15 ± 1.95	8.85 ± 0.11	2.95 ± 0.21	17.05 ± 0.16
SB-10	28.25 ± 1.21	8.22 ± 0.66	7.65 ± 0.21	29.05 ± 1.80	7.16 ± 0.09	3.16 ± 0.33	16.52 ± 0.21
SB-20	15.45 ± 1.22	24.27 ± 0.73	6.59 ± 0.14	27.07 ± 1.62	7.89 ± 0.07	3.26 ± 0.46	15.47 ± 0.12
SB-30	8.03 ± 1.94	32.74 ± 0.11	7.23 ± 0.11	28.68 ± 2.10	7.05 ± 0.07	4.04 ± 0.43	12.23 ± 0.32
SB-40	42.32 ± 1.88	–	5.55 ± 0.67	25.82 ± 2.11	9.74 ± 0.05	3.41 ± 0.14	13.16 ± 0.27

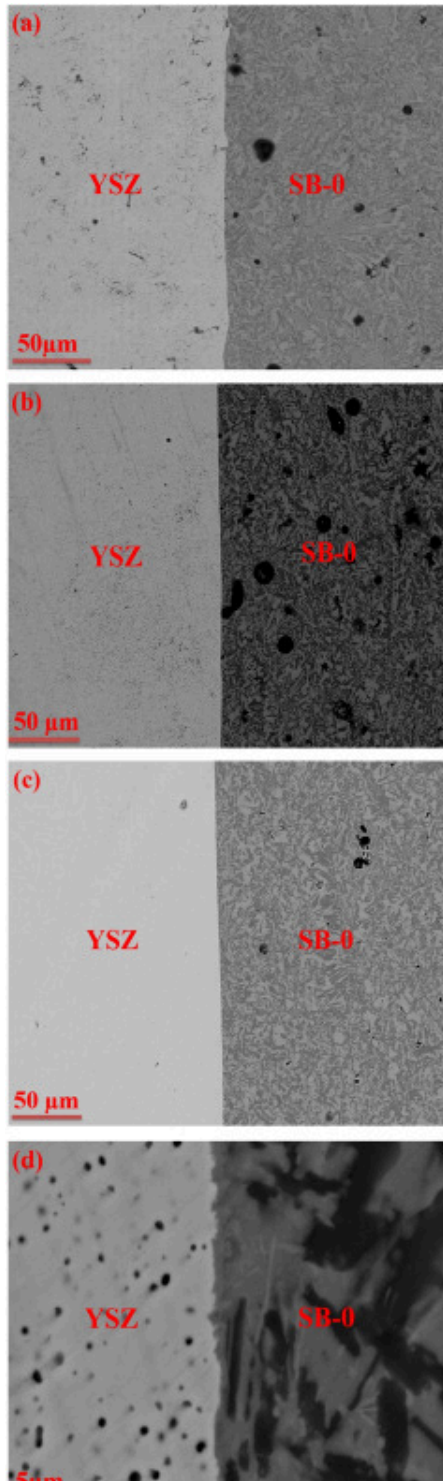
YSZ pellets (10×10×3 mm³) are formed using hydraulic press by applying 12 kN cm^{−2} pressure. These pellets are sintered at 1200 °C for 12 h in a high temperature furnace in air. These sintered pellets are monophasic and exhibited the cubic phase. The as prepared glass powder is used to make the slurry in poly vinyl chloride (PVA) and applied on the sintered pellets of YSZ. The diffusion couple of YSZ/glass sealant are heat treated above softening temperature (T_s) of glass sealants [27]. It is heat treated at 900 °C for 45 min so that PVA should evaporate and glass sealant makes some bonds with YSZ. After that, the diffusion couples underwent heat treatment for 1, 10, 100, and 1000 h at 900 °C in presence of air at atmospheric pressure.

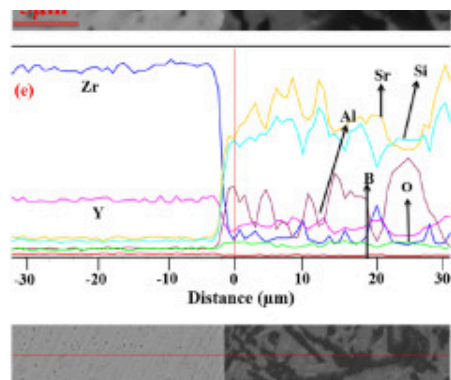
After, heat treatment, for different heat treatment durations, the diffusion couple is mechanically polished on various grad emery paper followed by diamond polishing using 1 μm particle size diamond paste followed by dilute HF etching of the surface to conduct the field emission scanning electron microscope (FESEM) study. A high-resolution field emission scanning electron microscope (FESEM: Sigma500, Zeiss, Germany) operating at 20 kV was used for morphological analysis. Gold coating (Rotatory pumped coater: Quorum, Q 150R ES plus) was done at 20 mA for 60 s on carbon tape in the presence of argon gas to avoid a charging effect during FESEM observation. The gold layer has a thickness of a few nanometers. An energy dispersive spectroscopy (EDS) was used to perform chemical analysis on glass sealants and diffusion couples at different places to check the composition and diffusion of various elements from sealants towards YSZ and vice-versa with respect to the thermal cycle. A different glass-YSZ diffusion couple was mechanically ground in to powder form using mortar and pestle. The fine powder sample was then used to study the crystalline phases formation during interaction study. For this purpose, the X-ray diffraction technique was used. The experiment was performed on Rigaku-SmartLab SE X-ray diffraction (XRD: SmartLab SE, Rigaku Corporation, Japan), and the software used for XRD measurement is SmartLab Studio II. The step size during the experiment was 0.01° and the angular range was between 10° – 80° with a scan rate of 5° min^{-1} . The X-ray source was operated by a 30-mA current and 40 kV accelerating voltage. PAN analytical X'Pert high score plus system with Cu- $K\alpha$ radiations ($\lambda = 1.5406 \text{ \AA}$) was used to investigate the nature of phases. In addition to FESEM, the analytical composition of glasses and line scan of the diffusion of elements within the diffusion couples were investigated using Energy Dispersive X-ray Spectroscopy (EDS: XFlash 6130, Quantax) in low vacuum mode at 23°C and 44–51 % relative humidity. To get an idea about the hardness of formed interface, the hardness was measured in the transverse direction to the diffusion couples at YSZ, interface and glass sealants side. The Hardness was measured by using Vicker's microhardness tester (model name: Mitutoyo) using a diamond indenter with a pyramidal base of the apical angle of 136° , by applying a force of 2.94 N with an indentation span of 15 s. The average value of hardness is taken by performing experiment at three places in YSZ/interface/glass sealants.

3. Results and discussion

3.1. Microstructural study

[Fig. 1\(a\)](#) shows the FESEM micrograph of SB-0/YSZ diffusion couple heat-treated at 900 °C for 10 h in air. Excellent adherence is shown by YSZ electrolyte and SB-0 glass sealant at low magnification ([Fig. 1\(a\)](#)). On the other hand, residual porosity is also observed in the glass sealant side. However, it is far from the interface. It may be associated with nucleation growth and formation of the crystalline solids in the glass sealants. Similar results have been observed during the interfacial study between glass sealants and $\text{SrZr}_{0.85}\text{Cu}_{0.15}\text{O}_{3-8}$ cathode [[29,30](#)]. The SB-0/YSZ couple interface, after heat treatment for 100 h at 900 °C is depicted in [Fig. 1\(b\)](#). The interface is still intact and free of any cracks and delamination. However, on the glass side, some porosity was visible. This could be because the glasses may have developed some crystalline phases. High gas tightness and a hermetic seal structure is predicted by the smooth and intact interface for the sealant [[31](#)].





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Fig. 1. FE-SEM micrographs SB-0/YSZ diffusion couple at 900 °C for heat duration (a) 10 h, (b) 100 h, (c) 1000 h, (d) magnified image of 1000 h heat duration, (e) Line profile of for 1000 h duration.

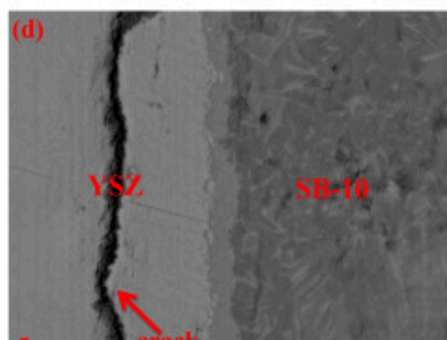
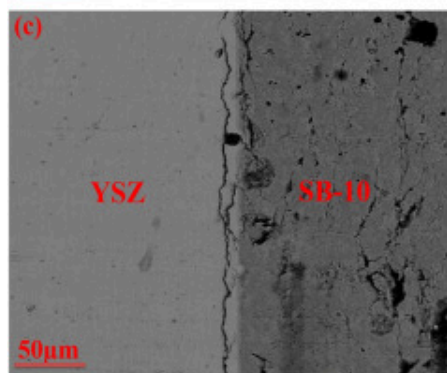
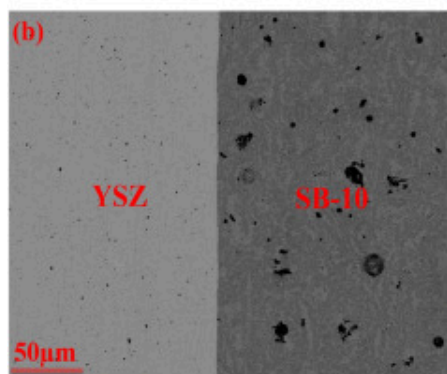
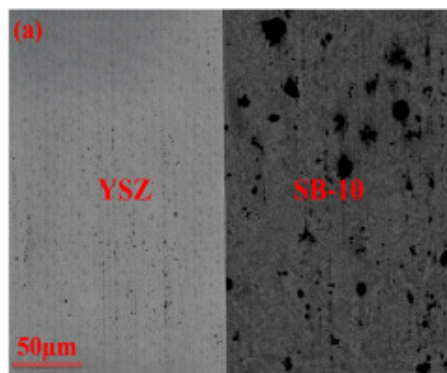
Even after prolonged heat treatment of 1000 h at 900 °C, the interface between SB-0/YSZ diffusion couple has shown good adherence (Fig. 1(c)). The SB-0 seal has shown good bonding characteristics and the absence of delamination even after prolonged heat-treatment for 1000 h. The seal region has low closed porosity, suggesting that this sealant would offer high gas tightness and a hermetic interface structure. Fig. 1(d) also displays higher magnification FESEM micrographs of the SB-0/YSZ interface (1000 h). Interface is found to be flawless and continuous.

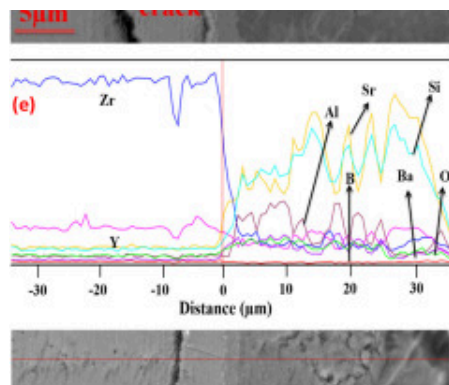
The literature reports that the heating rate is a critical factor in defining the microstructure and characteristics of the final product during glass powder sintering with concurrent crystallization [32]. The current glass sealant lacks visible particle boundaries and has pores that resemble spheres as shown in Fig. 1. This suggests that residual porosity is formed due to formation of bubbles during processing. Because of this dissolved gas and insoluble gases (such as air and water vapor) trapped in the original pores being released during thermal cycle [32]. EDS line scans were carried out on

diffusion couple heat treated for 1000 h at 900 °C as shown in [Fig. 1\(d\)](#) which confirmed the presence of the Si, Sr, and Al elements and their diffusion from glass sealant side to YSZ and Zr from YSZ to glass sealant side as shown in [Fig. 1\(e\)](#).

There has been no segregation or accumulation at the interface due to different ions diffusion from the YSZ to glass sealants or vice versa. To know the crystalline phases formation during interaction study, the XRD patterns were also taken after exposed the samples at various heat treatment durations as discussed and correlated with microstructural analysis in [Section 3.2](#).

[Fig. 2\(a–c\)](#) gives the interfacial microstructure for the SB-10/YSZ diffusion couple heat-treated at 900 °C for 10 h, 100 h, and 1000 h durations, respectively. The magnified FESEM micrograph for the thermally cycled SB-10/YSZ diffusion couple at 900 °C for a 1000 h heat treatment period is displayed in [Fig. 2\(d\)](#). The intact and pore-free interface is depicted in [Fig. 2\(a–c\)](#) with the formation of crack around the interface in 1000 h FESEM figure. The glass sealant exhibits dendritic type microstructure and is found to be very uniform distribution throughout the sealant [\[33\]](#). However, the glass sealant (SB-10) side exhibits more porosity as compared to the SB-0/YSZ sample. When the heat treatment duration increases from 10 h to 100 h, the porosity is reduced in glass side. It means glass sealant has self-healing tendency as also reported in BaO content glass sealants [\[\[34\], \[35\], \[36\]\]](#).





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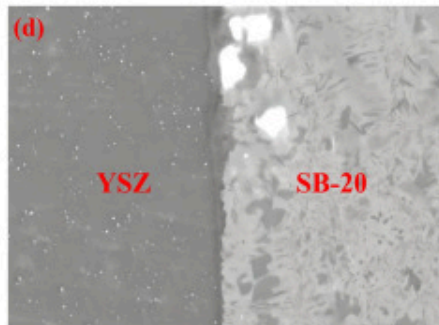
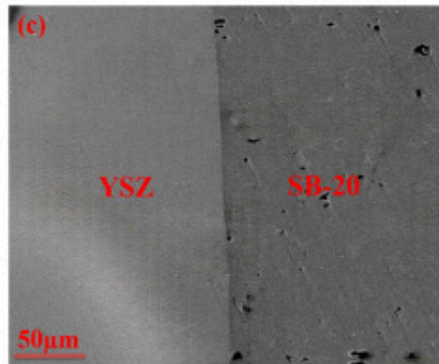
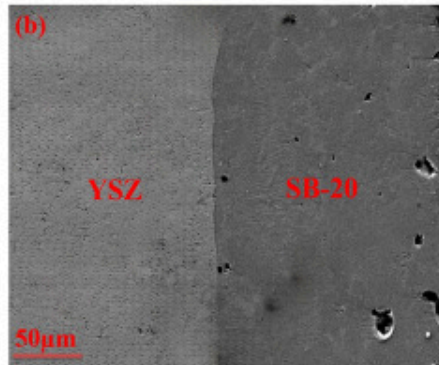
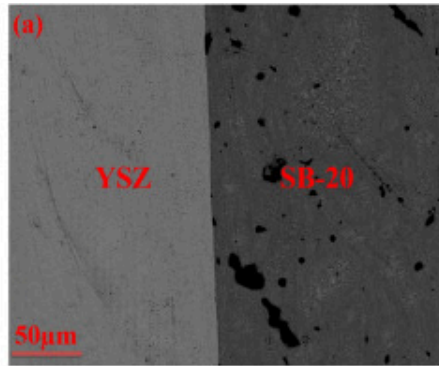
Fig. 2. FE-SEM micrographs SB-10/YSZ diffusion couple at 900 °C for heat treatment duration (a) 10 h, (b) 100 h, (c) 1000 h, (d) magnified image of 1000 h heat treatment duration, (e) Line profile of different elements after 1000 h time duration.

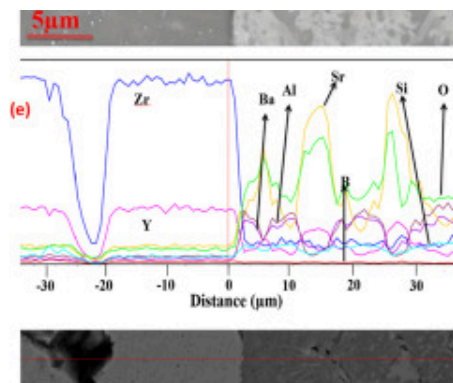
According to Fig. 2(e), EDS line scan conducted on the diffusion couple heat treated for 1000 h at 900 °C did not indicate any yttrium diffusion from YSZ side to the glass-sealant side. However, the presence of Sr, Ba, Si, and Zr are observed at the interface. No elemental segregation is observed at the interface. Zirconium exhibits the highest concentration in the glass sealant side, according to the diffusion pattern, because it diffuses from the electrolyte side to the glass sealant side, followed by diffusion of silicon and strontium from the glass side. It seems that the presence of both the modifiers (SrO and BaO) in glass sealants not only enhance the diffusion of Zr from YSZ side but also formed the crack in YSZ side. It could be associated with detrimental crystalline phases formation as discussed in Section 3.2.

For every element, there are oscillations in the concentration profiles. In the glass region, more oscillations are observed in the Sr, Si, and Zr spikes. This could be due to distinct elemental interactions; low concentrations of glass elements and bigger pores and cracks are present in the sealants [37]. The prominent feature is the continuous cracks is

observed in the vicinity of the interface at YSZ side after exposing at 1000 °C. It is possible due to CTE mismatch between different phases formed around the interface at 1000 h. This particular glass sealant has formed the $\text{Sr}_3\text{Zr}_2\text{O}_7$ crystalline phase in addition to other crystalline phase as discussed in XRD sections. The formation of these crystalline phase and their volume fractions might be responsible for the crack nearby interface in this particular glass sealants [38,39].

FESEM image of the interface between YSZ/SB-20 glass sealant after heat treatment at 900 °C for 10 h in air ambient is shown in Fig. 3(a). Glass and YSZ show a strong bond, because of the presence of continuous interface. Additionally, an extended crystalline microstructure is seen at the interface. The FESEM micrographs of the SB-20/YSZ diffusion couple after heat treated for 100 h at 900 °C are presented in Fig. 3(b). The interface is again found to be completely uniform, testifying the homogeneity of the prepared glass and a vermicular phase separated structure is evident having less porosity than SB-0/YSZ and SB-10/YSZ diffusion couple [40].





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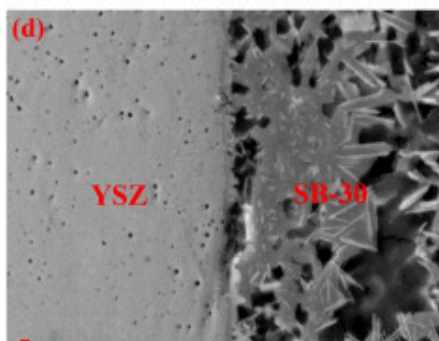
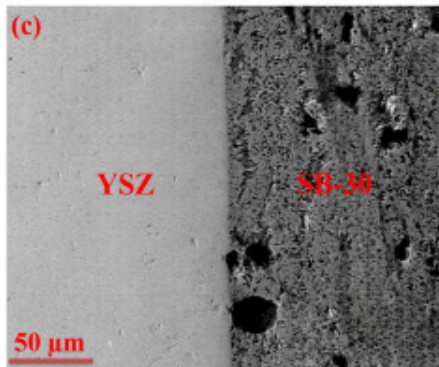
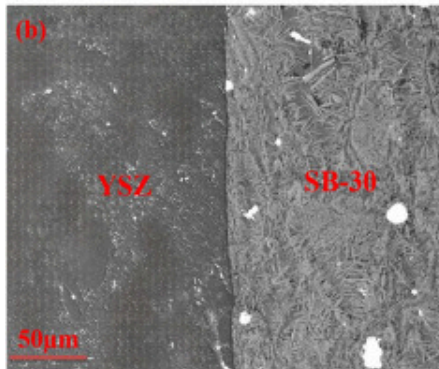
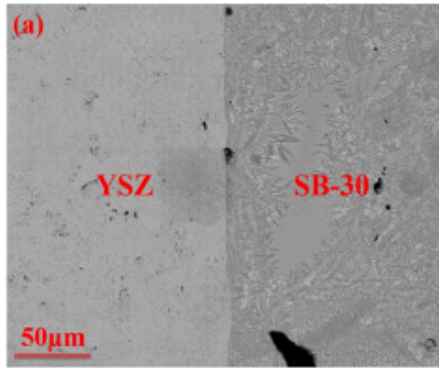
Fig. 3. FE-SEM micrographs SB-20/YSZ diffusion couple at 900 °C for heat duration (a) 10 h, (b) 100 h, (c) 1000 h, (d) magnified image of 1000 h heat duration, (e) Line profile of 1000 h duration exposed diffusion couple.

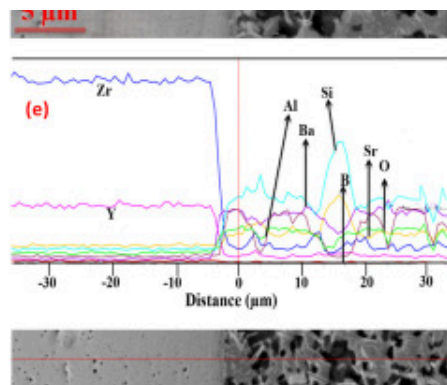
Upon heat treating at 900 °C for 1000 h, interface is still intact and smooth except some porosity at the interface. The highly interconnected nature of the two phases clearly indicates the spinodal decomposition and phase separation in magnified FESEM as shown in Fig. 3(d). Fig. 3(d) shows the magnified FESEM micrograph for the thermally cycled SB-20/YSZ diffusion couple at 900 °C for 1000 h heat treatment duration. No significant reaction that might have led to the formation of a new compound happened at the interface. However, a sequence of extraordinarily large pores developed throughout the boundary of the interface particularly in the glass sealant side. It is evident that BaO addition enhance the crystallisation in glass sealant along with porosity in the glass sealant side. The number of crystalline phases is also increased as discussed in Section 3.2.

There was a population of smaller pores in addition to the bigger one. The interface revealed that there were no cracks in the joint area [40]. As seen in Fig. 3(e), EDS line scan conducted on the diffusion couple heat treated for 1000 h at 900 °C did not indicate any

yttrium diffusion from YSZ into the glass-ceramic. Again, there was no sign of any elemental diffusion or penetration into the YSZ side of the glass. It showed that good bonding with YSZ is caused by interdiffusion of Zr from YSZ side to glass side and of Ba, Si, and Al from glass side to YSZ side.

FESEM image of the interface between YSZ/SB-30 glass after heat treatment at 900 °C for 10 h is shown in [Fig. 4\(a\)](#). For the SB-30/YSZ diffusion couple, there is excellent adhesion along the joint region as indicated from [Fig. 4\(a\)](#). The sealing glass is tightly bonded to the solid electrolyte, and there was no interfacial delamination at the interface. In the case of 100 h heat treatment duration ([Fig. 4\(b\)](#)), the interface has shown some pores though the glass composite exhibits fairly homogenous microstructure without any severe corrosion effects at the interface [[41,42](#)].





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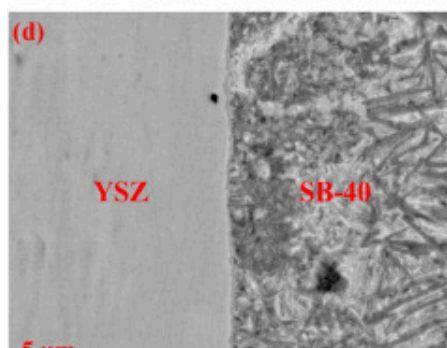
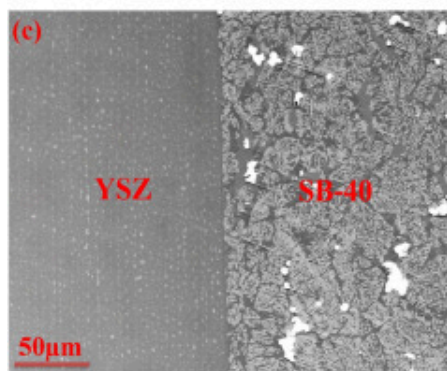
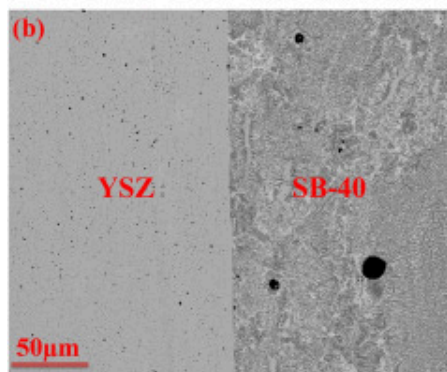
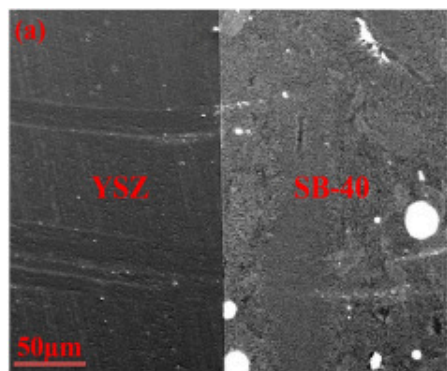
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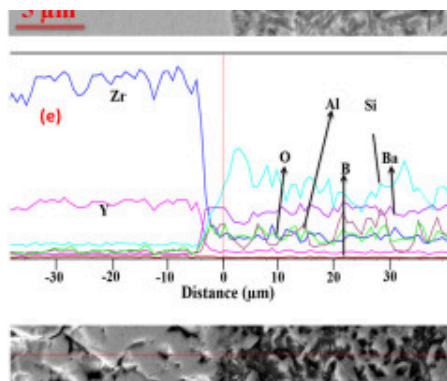
Fig. 4. FE-SEM micrographs SB-30/YSZ diffusion couple at 900 °C for heat duration (a) 10 h, (b) 100 h, (c) 1000 h, (d) magnified image of 1000 h heat duration, (e) Line profile of 1000 h duration exposed couple.

Additionally, the diffusion couple that has been heat-treated for a 1000 h shows the formation of pores at the joint area (Fig. 4(c)) as discussed in the XRD section. The size of these pores is greater in size as compared to the pores formed in SB-20 diffusion couple when heat treated at 1000 h. It is further shown by line mapping done on diffusion couple heat treated for 1000 h at 900 °C in Fig. 4(e) that elemental segregation is fully prevented around the sealing region. Although there isn't any apparent zirconate formation, there has been some subsequent alkaline earth metal ion diffusion. Only, zirconium has shown diffusion in almost all the diffusion couples.

Fig. 5(a) shows the FE-SEM micrograph of YSZ/SB-40 heat-treated at 900 °C for 10 h in air. Fig. 5(b) presents a back-scattered electron micrograph of a pure SB-40/YSZ interface heat-treated for 100 h. The remaining glass has two crystalline phases (darker phase). After 100 h of heat treatment, there are still no pores or cracks in the interface. The remaining glass has two crystalline phases (darker phase). After 100 h of heat treatment, there are still no pores or cracks in the interface. Additionally, the diffusion couple that

has been heat-treated for a 1000 h shows good bonding qualities with no delamination at the joint area ([Fig. 5\(c\)](#)). Moreover, detrimental crystalline phase is not formed in this diffusion couple as discussed in [Section 3.2](#).





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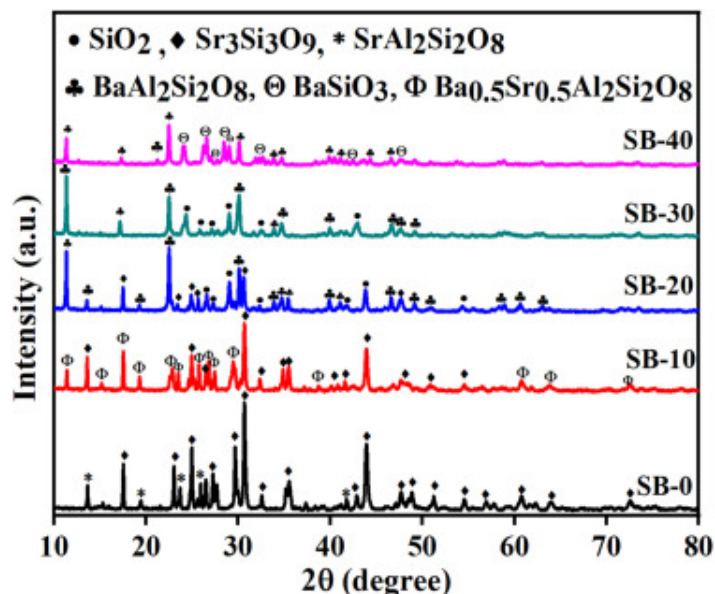
Fig. 5. FE-SEM micrographs SB-40/YSZ diffusion couple at 900 °C for heat duration (a) 10 h, (b) 100 h, (c) 1000 h, (d) magnified image of 1000 h heat, (e) Line profile of 1000 h duration exposed diffusion couple.

The experimental results point to a strong interfacial adhesion between YSZ and glass. Specifically, upon cross section and polishing, very good adhesion is seen, and continuous interfaces devoid of fractures or gaps are visible [43].

The EDS line scan performed at magnified Fig. 5(d) of 1000 h heat treatment duration detected Ba, Si and O in these precipitates, but with different relative intensities as shown in Fig. 5(e). However, the formation of an interfacial reaction layer is the primary observational feature [25]. Zr, Ba, Al, Si, and O were found at the contact with the YSZ substrate by the line mapping analysis, indicating the formation of $\text{BaAl}_2\text{Si}_2\text{O}_8$ and Ba_2SiO_5 . Zr, Ba, and O were also found by the EDS line mapping study, indicating the formation of BaZrO_3 . XRD was used to confirm the existence of this zirconate. Eventually, a small peak of Zr was detected by EDS analysis on the glassy phase, indicating that ZrO_2 was somewhat soluble in the glass sealant as also reported earlier [44].

3.2. XRD analysis of glass sealants after exposed for different durations

Fig. 6 shows XRD of as-prepared glasses after 500 h heating at 900 °C. SB-0 glass exhibits the formation of $\text{Sr}_3\text{Si}_3\text{O}_9$ [ICDD-010,752,281] and $\text{SrAl}_2\text{Si}_2\text{O}_8$ [ICDD-000,100,015]. SB-10 glass shows formation of $\text{Ba}_{0.5}\text{Sr}_{0.5}\text{Al}_2\text{Si}_2\text{O}_8$ [ICDD-000,381,452] phase. SB-20 glass shows the formation of SiO_2 phase [ICDD-010,751,381], $\text{BaAl}_2\text{Si}_2\text{O}_8$ phase [ICDD-010,881,048] and minor phase of $\text{Sr}_3\text{Si}_3\text{O}_9$ [ICDD-010,752,281]. SB-30 glass shows the formation of SiO_2 phase [ICDD-010,751,381] and $\text{BaAl}_2\text{Si}_2\text{O}_8$ phase [ICDD-010,881,048]. SB-40 glass shows the formation of BaSiO_3 [ICDD-000,261,402] and $\text{BaAl}_2\text{Si}_2\text{O}_8$ phase [ICDD-010,881,048]. Interestingly, ZrO_2 contained crystalline phase is not formed it could be associated with lower amount of ZrO_2 (5 mol %) in the glass composition. However, it has been reported that ZrO_2 (10 wt %) in $\text{CaO-K}_2\text{O-Na}_2\text{O-BaO}$ silicate glass control the CTE with decrease stress and higher viscosity [45].

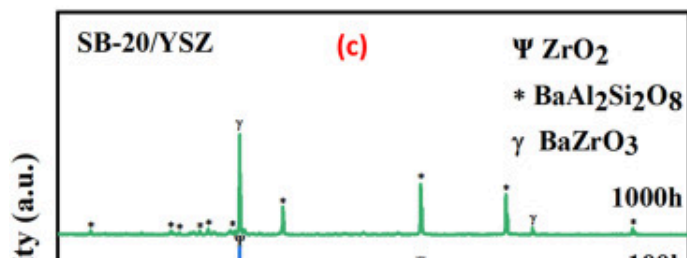
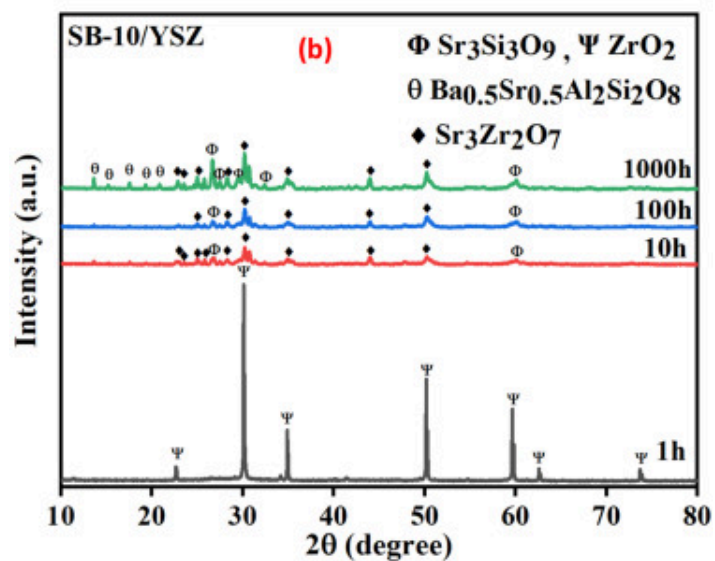
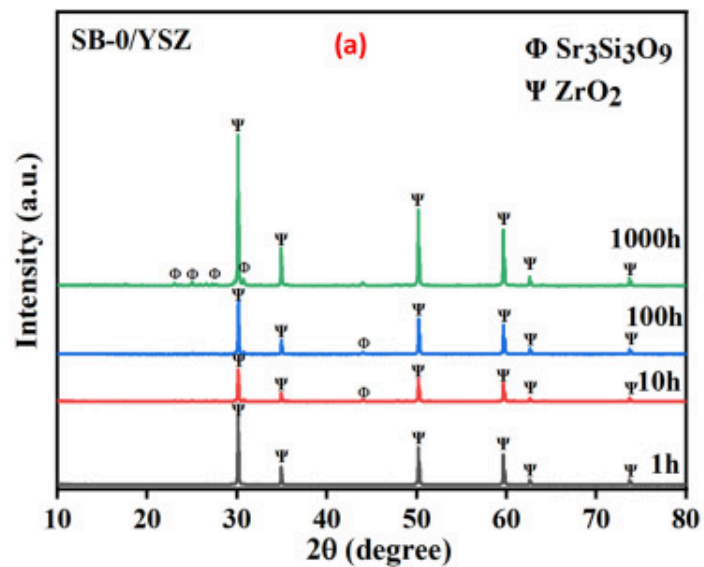


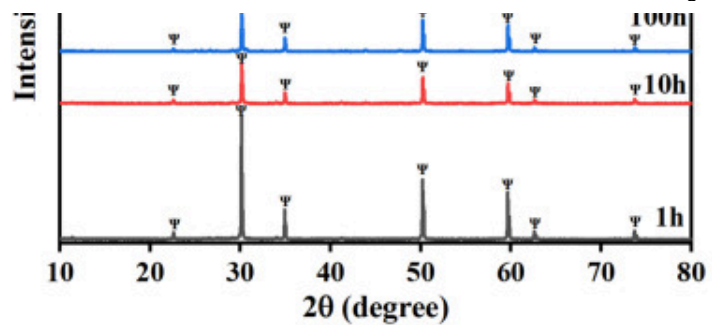
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Fig. 6. XRD of SB-0, SB-10, SB-20, SB-30 and SB-40 glasses heat treated for 500 h at 900 °C.

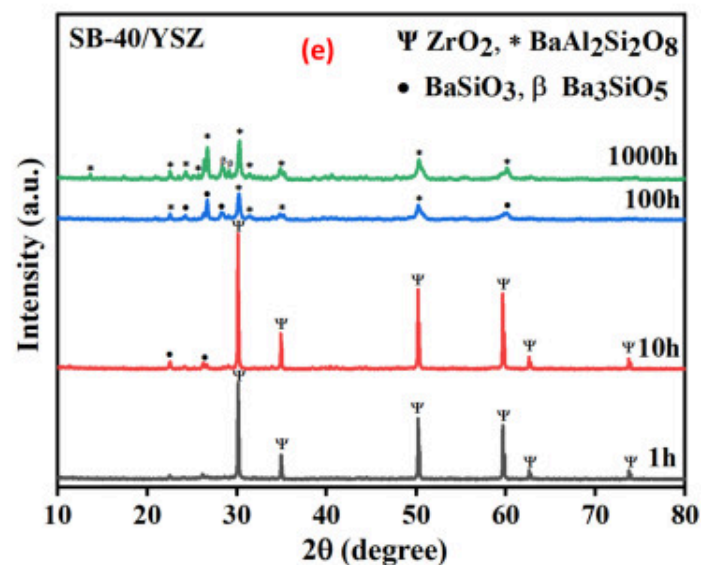
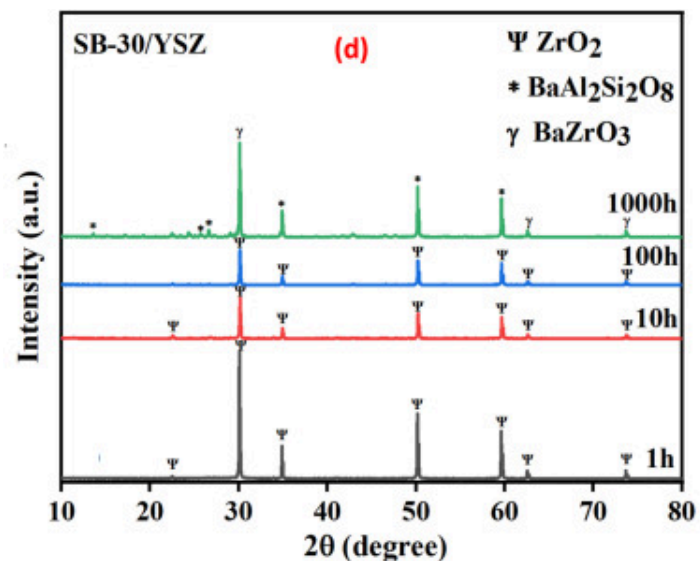
After exposure for different durations at 900 °C, X-ray diffraction pattern of coupled SB-0/YSZ, SB-10/YSZ, SB-20/YSZ, SB-30/YSZ, and SB-40/YSZ after heat-treatment at 900 °C for durations of 1, 10, 100 and 1000 h is shown in [Fig. 7](#)(a–c).





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Fig. 7. XRD of the chemical compatibility of (a) SB-0 glass with YSZ, (b) SB-10 glass with YSZ, (c) SB-20 glass with YSZ, (d) SB-30 glass with YSZ, (e) SB-40 glass with YSZ for 1 h, 10 h, 100 h and 1000 h at 900 °C, respectively.

The SB-0/YSZ couple shows the formation of monoclinic ZrO_2 phase at heat treatment duration of 1 h. As heat treatment duration is increased from 1 h to 10 h, a crystalline phase of $\text{Sr}_3\text{Si}_3\text{O}_9$ is formed along with ZrO_2 [ICDD-010,899,069] as shown in Fig. 7(a). But as the duration of heat treatment is increased to 100 h, volume fraction of $\text{Sr}_3\text{Si}_3\text{O}_9$ [ICDD-010,752,281] phase is increased. Similar results have been reported by other researchers in different glass compositions after the interaction with YSZ [29]. These results are in accordance with FESEM results because formation of both these phases is not proved to be detrimental for the working of SOFC hence there was no crack formation around the interface.

The heat-treatment of SB-10 glass/YSZ couple shows the formation of monoclinic ZrO_2 phase at heat treatment duration of 1 h as shown in Fig. 7(b). But as BaO with 10 mol % is added to this glass as compared to SB-0, major peaks is replaced from ZrO_2 [ICDD-010,899,069] to $\text{Sr}_3\text{Zr}_2\text{O}_7$ phase [ICDD-010,731,257] when the diffusion couple is heated for 10, 100 and 1000 h. The 10 and 100 h heat treatment duration also observed the formation of $\text{Sr}_3\text{Si}_3\text{O}_9$ phase [30]. This crystalline phase exhibit negative CTE below room temperature [46]. At higher temperature CTE of these type of phase $\text{Sr}_3\text{Ti}_2\text{O}_7$ exhibit $32 \times 10^{-6} / \text{K}$ at 627°C (10 GPa) [47]. Thus, the steep change in CTE due to $\text{Sr}_3\text{Zr}_2\text{O}_7$ might be responsible to develop crack near the interface SB-10/YSZ diffusion couple as observed in Fig. 1(c and d) in prolonged heat treatment duration. Probable $\text{Ba}_{0.5}\text{Sr}_{0.5}\text{Al}_2\text{Si}_2\text{O}_8$ phase formation increase the crack formation.

In case of 1000 h heat treatment duration, the main XRD peaks of the $\text{Sr}_3\text{Zr}_2\text{O}_7$ phase overlap with those from the $\text{Sr}_3\text{Si}_3\text{O}_9$ phase. The reference spectrum for $\text{Sr}_3\text{Zr}_2\text{O}_7$ also shows that the major peaks of this compound overlap with those associated with $\text{Ba}_{0.5}\text{Sr}_{0.5}\text{Al}_2\text{Si}_2\text{O}_8$ phase [ICDD-000,381,452]. The details of these phases are given in Table 1. This particular glass sealant shows the formation of continuous crack along the interface. Contrary to this, strontium zirconate phase has CTE around $(10-11 \times 10^{-6} / \text{K})$ [48]. $\text{Ba}_{0.5}\text{Sr}_{0.5}\text{Al}_2\text{Si}_2\text{O}_8$ phase has a very low CTE ($\sim 3-4 \times 10^{-6} / \text{K}$) [49]. This mismatch in CTE has led to formation of the crack at the interface. However, the CTE of as-prepared SB-10 glass was $10.53 \times 10^{-6} / \text{K}$ [27].

The heat-treatment of SB-20 glass/YSZ couple shows the formation of monoclinic ZrO_2 phase and it is only known phase up to heat treatment duration of 100 hour as shown in Fig. 7(c) and Table 2.

Table 2. Different phases with their volume fraction in YSZ/glass diffusion couple after heat treatment.

Diffusion Couple Code	Crystalline phase (volume fraction) at 1 h	Crystalline phase (volume fraction) at 10 h	Crystalline phase (volume fraction) at 100 h	Crystalline phase (volume fraction) at 1000 h
SB-0/YSZ	$\text{ZrO}_2(100)$	$\text{ZrO}_2(87)$	$\text{ZrO}_2(92)$	$\text{ZrO}_2(89)$
	–	$\text{Sr}_3\text{Si}_3\text{O}_9(13)$	$\text{Sr}_3\text{Si}_3\text{O}_9(8)$	$\text{Sr}_3\text{Si}_3\text{O}_9(11)$
SB-10/YSZ	$\text{ZrO}_2(100)$	$\text{Sr}_3\text{Zr}_2\text{O}_7(95)$	$\text{Sr}_3\text{Zr}_2\text{O}_7(92)$	$\text{Sr}_3\text{Zr}_2\text{O}_7(61)$
	–	$\text{Sr}_3\text{Si}_3\text{O}_9(5)$	$\text{Sr}_3\text{Si}_3\text{O}_9(8)$	$\text{Sr}_3\text{Si}_3\text{O}_9(27)$
	–	–	–	$\text{Ba}_{0.5}\text{Sr}_{0.5}\text{Al}_2\text{Si}_2\text{O}_8(12)$
SB-20/YSZ	$\text{ZrO}_2(100)$	$\text{ZrO}_2(100)$	$\text{ZrO}_2(100)$	–
	–	–	–	$\text{BaZrO}_3(23)$
	–	–	–	$\text{BaAl}_2\text{Si}_2\text{O}_8(77)$
SB-30/YSZ	$\text{ZrO}_2(100)$	$\text{ZrO}_2(100)$	$\text{ZrO}_2(100)$	–
	–	–	–	$\text{BaZrO}_3(39)$
	–	–	–	$\text{BaAl}_2\text{Si}_2\text{O}_8(61)$
SB-40/YSZ	$\text{ZrO}_2(100)$	$\text{ZrO}_2(88)$	$\text{ZrO}_2(59)$	$\text{BaZrO}_3(18)$
	–	$\text{BaSiO}_3(12)$	$\text{BaSiO}_3(35)$	–

Diffusion Couple Code	Crystalline phase (volume fraction) at 1 h	Crystalline phase (volume fraction) at 10 h	Crystalline phase (volume fraction) at 100 h	Crystalline phase (volume fraction) at 1000 h
	–	–	BaAl ₂ Si ₂ O ₈ (6)	BaAl ₂ Si ₂ O ₈ (71) Ba ₃ SiO ₅ (11)

When the diffusion couple is heat treated for 1000 h, BaZrO₃ [ICDD-000,060,399] and monoclinic BaAl₂Si₂O₈ [ICDD-000,381,450] phases are formed having CTE around ($\sim 7.13 \times 10^{-6}/\text{K}$) and ($\sim 7.86 \times 10^{-6}/\text{K}$) respectively [49,50]. Basically, ZrO₂ the diffusion of Ba and Zr leads to conversion of ZrO₂ into BaZrO₃. On the other hand, Al₂O₃ picked up from the alumina crucible during glass melting and is diffused into the glass network. It is discovered that the volume fraction of these phases increases as the heat treatment time is increased from 100 to 1000 h. The formation of alumina related (BaAl₂Si₂O₈) phase is associated due to diffusion of Al from Al₂O₃ crucible [39]. The X-ray diffractograms show more prominent crystalline peaks at higher heat-treated diffusion couple, which suggests that the crystallinity in glass sealant is increasing with heat treatment duration. The initial formation of pores on the glass side is due to CTE mismatch between the glass and the YSZ. But at 1000 h heat treatment duration, the formation of pores is formed due to formation of BaZrO₃ phase around the interface. Since this phase (BaZrO₃) exhibits low CTE as compared to YSZ. Even the as-prepared SB-30 glass exhibit lowest CTE i.e., $7.75 \times 10^{-6}/\text{K}$ in all the glasses [27].

Fig. 7(d) is shown the XRD patterns of SB-30/YSZ diffusion couple heat treated for 1 h, 10 h, 100 h and 1000 h at 900 °C. The ZrO₂ [ICDD-010,899,069] that precipitated from the YSZ electrolyte side is the only major phase that developed up to a 100 h heat treatment duration. However, as the heat treatment duration is increased upto 1000 h, two new crystalline phases are also formed as also observed in SB-20/YSZ diffusion couple. The monocelsian (BaAl₂Si₂O₈) phase [ICDD-000,381,450] at high temperatures is the primary

phase that typically crystallizes first in SB-30 glass [19]. The volume fraction of BaZrO₃ [ICDD-000,060,399] having CTE around $\sim 7.9 \times 10^{-6}/\text{K}$ in this glass is very high than SB-20 glass at 1000 h heat treatment duration, which is the reason of bigger pores around the interface at 1000 h heat treated diffusion couple due to CTE mismatch between the phases monocelsian BaAl₂Si₂O₈ and BaZrO₃ phase [48].

Fig. 7(e) shows the XRD pattern of SB-40/YSZ diffusion couple heat treated for 1 h, 10 h, 100 h and 1000 h at 900 °C. The glass diffusion couple after heat treatment up to 100 h shows the formation of BaSiO₃ [ICDD-000,261,402] and monocelsian BaAl₂Si₂O₈ phase [ICDD-000,381,450] along with monoclinic ZrO₂ [ICDD-010,899,069]. The predominant crystalline component, barium silicate (BaSiO₃) (CTE $\sim 9.4\text{--}12.5 \times 10^{-6}/\text{K}$) [50], is still present and was formed during the first heat treatment phase. In actuality, the overall amount of this phase steadily rises over time when maintained at 900 °C. The more stable monoclinic structure starts to take the place of the metastable monocelsian ingredient at 100 h. The substantial drop in CTE, for the thermally aged bulk glass-ceramic is due to the formation of this phase. The overall result of continual exposure to high temperatures is a decrease in the CTE of the glass-ceramic. On the other hand, the low intensity of crystalline reflections suggests that the samples are not properly crystallized [39]. However, when the diffusion couple is heat treated for 1000 h at 900 °C, it showed comparatively good crystallization behaviour. XRD patterns of this diffusion couple after controlled heat treatment at 900 °C for 1000 h reveal the formation of Ba₂SiO₅ [ICDD-000,260,180], and BaAl₂Si₂O₈ [ICDD-000,381,450] along with formation of BaZrO₃ [ICDD-000,060,399] having CTE around $14.1 \times 10^{-6}/\text{K}$, $7.8 \times 10^{-6}/\text{K}$ and $7.13 \times 10^{-6}/\text{K}$ respectively [[49], [50], [51]]. The crystalline phase BaAl₂Si₂O₈ is detrimental for sealing point of view since it has higher CTE than required range for SOFC application. Contrary to this, another two crystalline phases i.e., Ba₂SiO₅ and BaZrO₃ exhibit the CTE in the required range of SOFC applications. Haishen et al. reported that the formation of hexagonal BaAl₂Si₂O₈ phase and its transformation in monoclinic BaAl₂Si₂O₈ decreases the CTE of glass sealants [52].

3.3. Diffusion constant and atomic fraction of diffused species

From the diffusion profile of the diffusing cations the diffusion coefficient is evaluated by using the following relation [53].

$$\frac{c(x,t)}{c_0} = 1 - \operatorname{erf} \frac{x}{2\sqrt{Dt}} \quad (1)$$

where $c(x,t)$ is the concentration of ions at the distance, x , from the interface for a given diffusion time period (t). The term c_0 is the initial concentration of ions and D the diffusion coefficient. The term erf stands for error function and its values vary from 0 to 1 for varying values of $x^2/4Dt$ between 0 and 3. The diffusion coefficient of Zr, Sr, Si and Ba in the sealants of the diffusion couples are summarized in Table 3.

Table 3. Diffusion constant for yttrium, strontium, silicon and barium in the SB-0/YSZ, SB-10/YSZ, SB-20/YSZ, SB-30/YSZ and SB-40/YSZ diffusion couples respectively. The values were determined from the diffusion across the interface in different time durations of heat treatment of diffusion couples at 900 °C.

Diffusion Couple Code	Time duration of heat treatment duration (h)	Zr diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})	Sr diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})	Si diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})	Ba diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})
SB-0/YSZ	10	0.017	0.0035	0.025	NA
	100	0.003	0.00035	0.00035	NA
	1000	0.0015	0.000035	0.000035	NA
SB-10/YSZ	10	0.72	0.0035	0.0035	0.0035
	100	0.044	0.00035	0.00035	0.00035
	1000	0.002	0.000035	0.00077	0.000035
SB-20/YSZ	10	0.63	0.00035	0.0077	0.00035

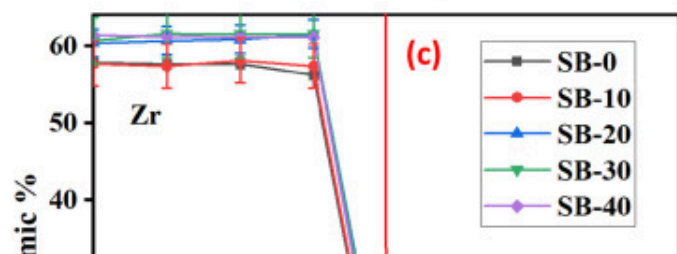
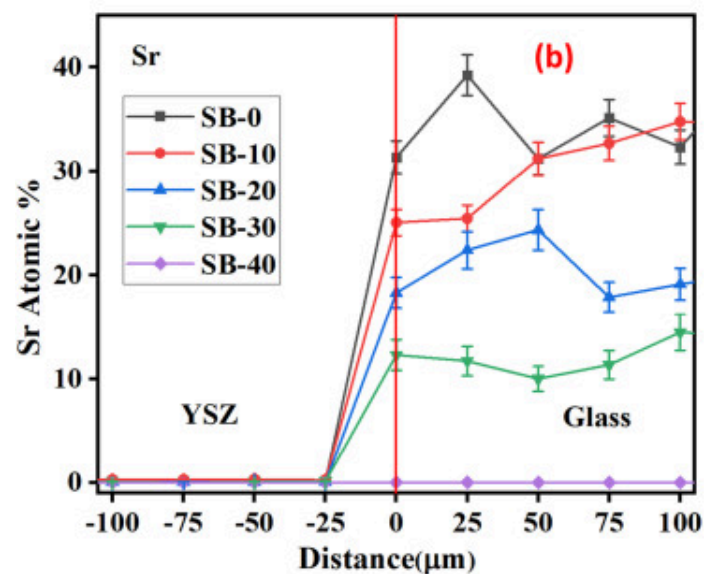
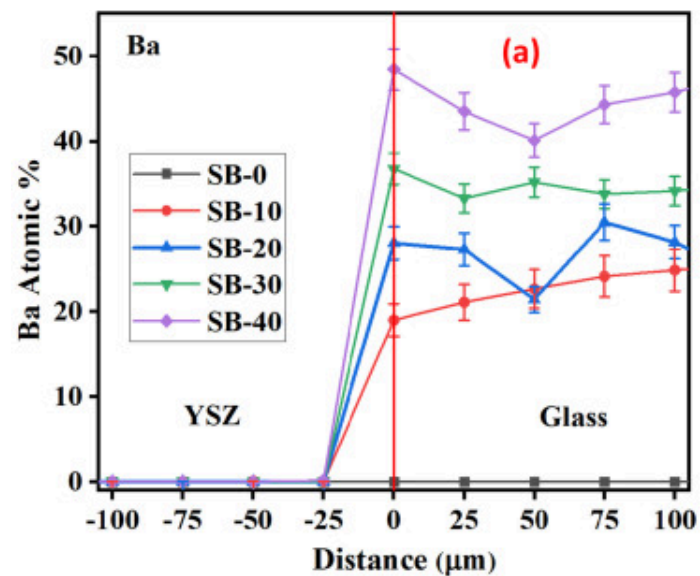
Diffusion Couple Code	Time duration of heat treatment duration (h)	Zr diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})	Sr diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})	Si diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})	Ba diffusion constant Values ($\times 10^{-12}$) (m^2s^{-1})
SB-30/YSZ	100	0.20	0.000035	0.000035	0.000035
	1000	0.04	0.0035	0.077	0.0035
	10	1.38	0.0035	0.0035	0.0035
	100	0.053	0.00035	0.00035	0.00035
	1000	0.0045	0.00030	0.000035	0.000035
SB-40/YSZ	10	0.25	NA	0.077	0.0035
	100	0.015	NA	0.0077	0.00035
	1000	0.0045	NA	0.000035	0.000035

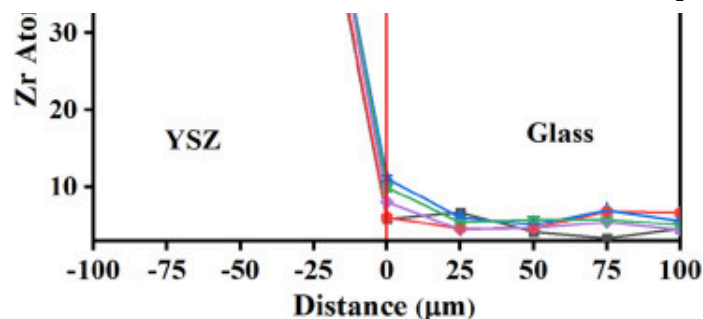
The diffusion coefficient of Zr^{4+} , Sr^{2+} , Si^{2+} and Ba^{2+} ions at the interface of the diffusion couple are summarized in [Table 3](#). Zirconium diffuses quickly toward glass sealant side of the interface at first because there is a greater concentration difference between the YSZ and the glass seal. Moreover, the ionic radius of Zr^{4+} is very small as compared to Y^{3+} , which sustains its fast diffusion kinetics [54]. Y^{3+} diffusion was very low because of its low concentration on YSZ side and is not shown in the [Table 3](#). As the heat treatment duration increases, the rate of diffusion of ions has decreased for all diffusion couples. It is well represented by many researchers [55,56]. Moreover, if it became constant with respect to time duration it is considered good for the sealing material [55]. The gradient in concentration may help to explain this phenomenon. Diffusion first occurs quickly because there is a significant concentration differential on both sides of the interface. However, the diffusion kinetics slow down as the concentration equilibrium is reached [56]. The value of diffusion constant of Zr^{4+} from YSZ side to glass side is more as

compared to diffusion constant values of Sr^{2+} , Si^{4+} and Ba^{2+} from glass side to YSZ side in all glasses for heat treatment duration of 10 h at 900 °C.

Zr is fastest diffusion is observed for SB-40/YSZ diffusion couple followed by SB-10/YSZ and SB-0/YSZ in 10 h of heat treatment. The diffusion of Zr is increased by BaO in the glass instead of SrO. Whereas, both the modifiers are present in glass sealants, the competition among both the modifiers slow down the diffusion. Sr diffusion is decreased with heat treatment durations. However, it is followed the similar trend in all the diffusion couple.

In case of Si the highest diffusion constant is observed for SB-20/YSZ. SB-30/YSZ diffusion couple, however, it is faster than the Sr and slower than the Zr whereas in case of Ba is follow the similar trend as observed for Sr in all the diffusion couple. Fig. 8 indicates the variation in the concentration of Sr, Zr and Ba in diffusion couples close to the interface, which are heat treated for 1000 h at 900 °C. The concentration of Zr decreases with increase in distance as we move from electrolyte side to the glass side (as shown in Fig. 8(c)) indicating a decrease in the diffusion of Zr into the glasses as the concentration of BaO is increased in glasses form SB-0 to SB-40. The concentration of Zr is found to be maximum in case of SB-30/YSZ diffusion couple. This result is in accordance with FESEM and XRD results as maximum volume fraction of ZrO_2 phases are formed in SB-30 and pores are increasing as SB-30/YSZ diffusion couple is heat treated for 10 h to 1000 h [56].





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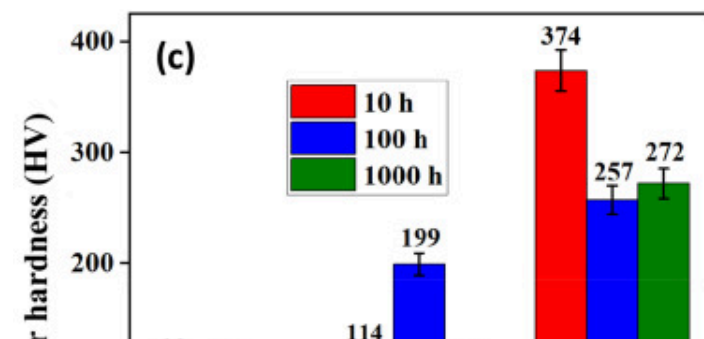
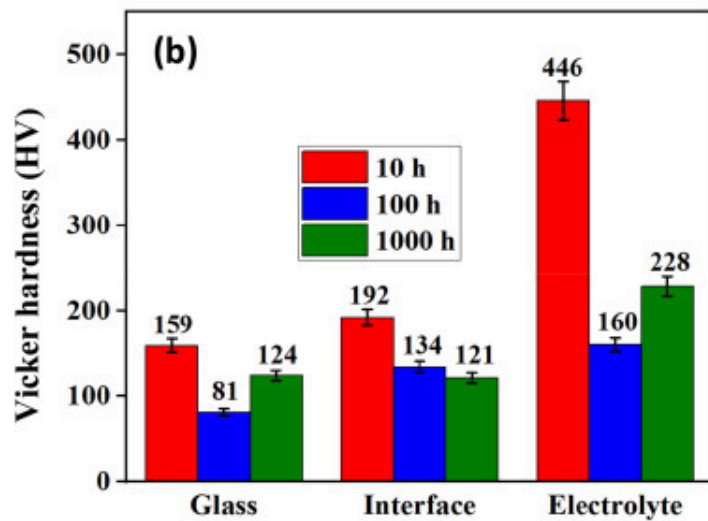
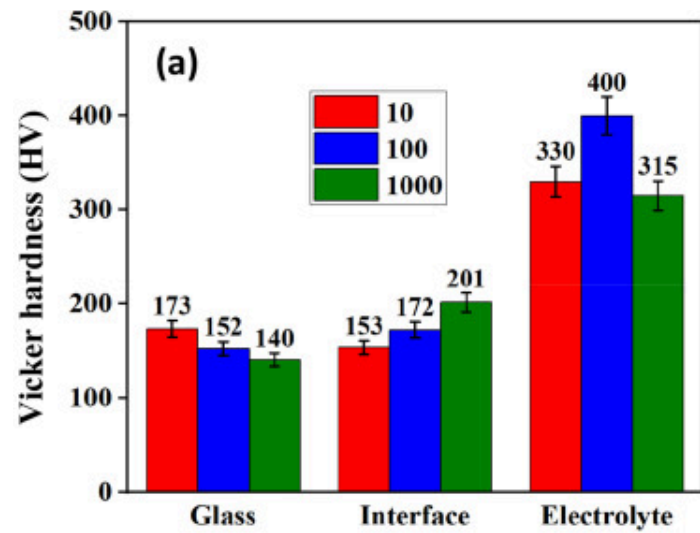
Fig. 8. Elemental concentration variation of (a) Ba (b) Sr (c) Zr in SB-0/YSZ, SB-10/YSZ, SB-20/YSZ, SB-30/YSZ and SB-40/YSZ with respect to diffusion couple distance.

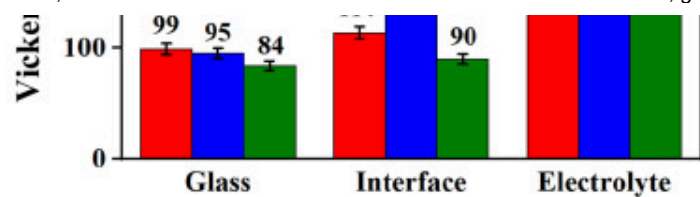
3.4. Vicker hardness

The microhardness test is performed on the polished surface of heat-treated diffusion couple for 10, 100, and 1000 h at 900 °C. The value of microhardness is determined by the following expression [57]:

$$H = 1.854 \times \frac{F}{D^2} \quad (2)$$

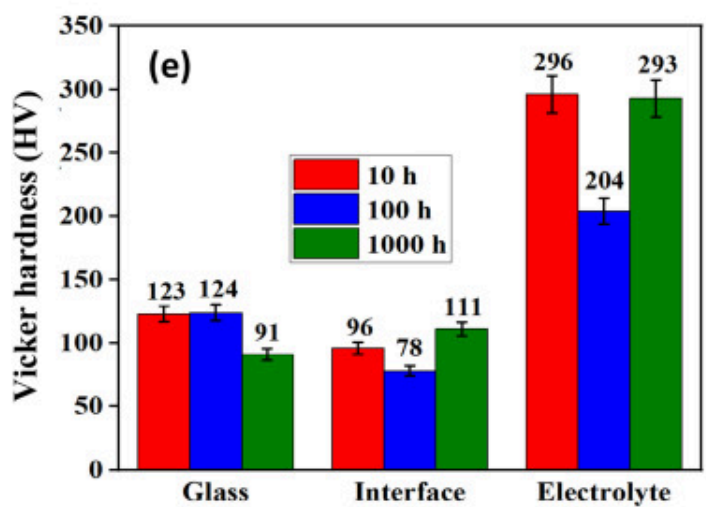
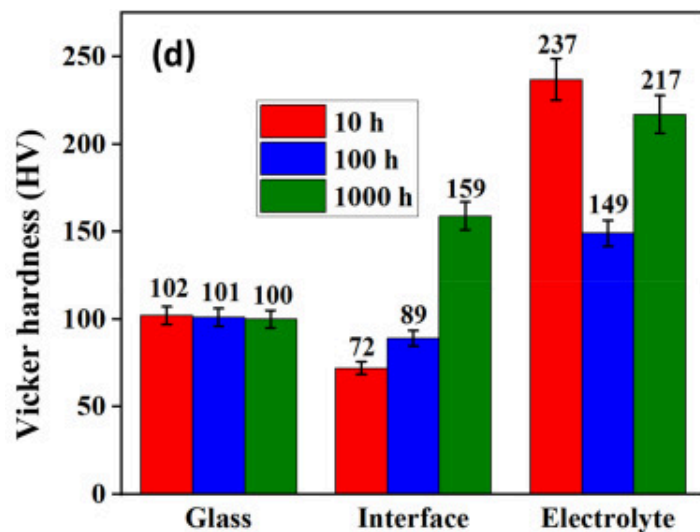
here F is the applied load and D is the diagonal length. Hardness values of all diffusion couples on either side of the center line are shown in Fig. 9. The maximum microhardness values were recorded for diffusion couple SB-0/YSZ for glass side is for 10 h heat treatment duration. The values of hardness at interface of all the samples was found to be in between of the values of glass side and YSZ. However, it has not followed any trend. The hardness value increases in direct proportion as temperature and time increases [56]. The minimum hardness value of 72 was recorded for the diffusion couple SB-20 at 1000 h heat treatment duration. The hardness values in the present study were maximum at the YSZ side and decreased as the distance is moved from YSZ side towards the glass side, which is in good agreement with the results reported in the literature [58,59].





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Fig. 9. Vicker hardness of different diffusion couples at glass sealant, interface and electrolyte (YSZ) after heat treatment duration for 10 h, 100 h and 1000 h at 900 °C of (a) SB-0/YSZ, (b) SB-10/YSZ, (c) SB-20/YSZ, (c) SB-30/YSZ, (d) SB-40/YSZ and (e) SB-40/YSZ.

On glass side, a decrease in the width of the diffused region was observed with the rise in the hardness values across the interface upto 100 h in case of all samples. The value increased for 1000 h heat treatment durations in case of SB-10, SB-30 and SB-40 diffusion couples, but it is found to be increasing as heat treatment duration is increased from 10 h to 1000 h in case of SB-0 and SB-20 glass diffusion couples. The annealing heat treatment procedure and processing temperature both correlate directly with increases in the hardness values.

However, the electrolyte (YSZ) side followed the exact opposite trend of the glass side. Conclusively, the Zr is the fast-moving ions from the YSZ side to the glass sealant side. On the glass sealant side Si is the fastest moving ions as compared to the Ba and Sr ions. It is obvious due to their smaller size in comparison to the other cations. These results are also verified by the X-ray diffraction and FESEM graphs of the diffusion couples. Since, most of the crystalline phases are contained the silicates or the Zr related crystalline phases. The highest hardness is observed in case of SB-10/YSZ. It is evident in FESEM of these diffusion couples. So, the strongest interface is formed in SB-10/YSZ for all the heat treatment duration.

4. Conclusion

SrO/BaO ratio affects the interface formation, growth and interfacial reaction with YSZ electrolyte. The addition of BaO in place of SrO increase the tendency of crystallization in the developed glass sealants. The smooth, pore free and intact interface with minimum reaction is observed where single alkaline earth metal oxide are in glass compositions i.e.

SB-0 and SB-40 glass sealant. The mixed modifiers are not only increasing the reactivity but also form some detrimental crystalline phase such as $\text{Sr}_3\text{Si}_3\text{O}_9$ which lead to the formation of crack at the interface in SB-10 glass sealant. The SB-0 glass sealant can be the best sealant where YSZ is being used as electrolyte in SOFC application. The highest diffusion has taken place by Zr and Si from YSZ and glass sealant side, respectively. The coherent interface is formed between SB-0/YSZ diffusion couple with highest hardness followed by SB-20/YSZ diffusion couple. The addition of ZrO_2 in glass sealant can reduce rate of diffusion of Zr from electrolyte YSZ side to glass sealant.

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CRediT authorship contribution statement

Trisha Walia: Writing – original draft, Resources, Methodology, Data curation. **K Singh:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Declaration of competing interest

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Data availability

Data will be made available on request.

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